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Problem 1

Show that

$$\frac{dp_\mu}{d\tau} = \frac{q}{c} u^\nu F_{\mu\nu}$$

leads to

$$\frac{d\vec{p}}{dt} = q \vec{E} + q \frac{\vec{v}}{c} \times \vec{B}$$

and

$$\frac{d}{dt} \left(\frac{U}{c} \right) = q \frac{\vec{v}}{c} \cdot \vec{E}.$$

Problem 2

We can replace $d\tau = \frac{dt}{\gamma}$

$$\gamma \frac{dp_\mu}{dt} = \frac{q}{c} u^\nu F_{\mu\nu} \quad (1)$$

$$= \frac{q}{c} (u^0 F_{\mu,0} + u^i F_{\mu,i}) \quad (2)$$

Because $F_{\mu,\nu} = -F_{\nu,\mu}$, we have

$$\gamma \frac{dp_\mu}{dt} = -\frac{q}{c} (u^0 F_{0,\mu} + u^i F_{i,\mu}) \quad (3)$$

We know that $u^0 F_{0,\mu} = -c F_{\mu}$, and we can relabel $F_\mu = \vec{E}$. This gives us

$$\gamma \frac{dp_\mu}{dt} = q \vec{E} - \frac{q}{c} u^i F_{\mu i} \quad (4)$$

1 Todo

Problem 2

Show that the Bianchi identity holds,

$$\epsilon^{\mu\nu\rho\sigma} \partial_\nu F_{\rho\sigma} = 0,$$

where

$$F_{\rho\sigma} = \partial_\rho A_\sigma - \partial_\sigma A_\rho.$$

$$= \epsilon^{\mu\nu\rho\sigma} \partial_\nu (\partial_\rho A_\sigma - \partial_\sigma A_\rho) \quad (5)$$

$$= \epsilon^{\mu\nu\rho\sigma} \partial_\nu \partial_\rho A_\sigma - \epsilon^{\mu\nu\rho\sigma} \partial_\nu \partial_\sigma A_\rho \quad (6)$$

$$(7)$$

2 TODO

Problem 3

Show that, under parity,

$$\vec{E}' = -\vec{E} \quad \vec{B}' = +\vec{B}$$

and under time reversal,

$$\vec{E}' = +\vec{E} \quad \vec{B}' = -\vec{B}.$$

$$E' = \Lambda_V^\mu E^\nu \quad (8)$$

3 TODO

Problem 4

Starting from $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, show that $\vec{B} = \vec{\nabla} \times \vec{A}$.

First, we can recognize that in matrix form, $\partial_\mu A_\nu = (\partial_\nu A_\mu)^T$.